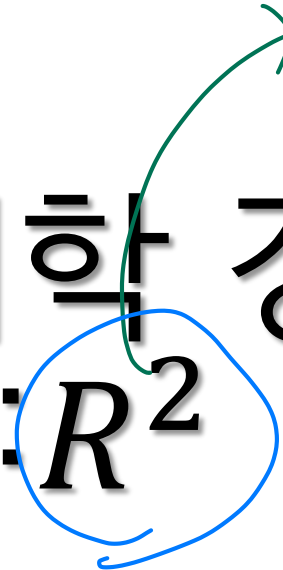


Goodness of fit

계량경제학 강의 10  
:  $R^2$



# 1. $R^2$ (R – square)

Workfile: UNTITLED

View Proc Object Sa

Range: 1 32 -- 32

Sample: 1 32 -- 32

☒ c

☒ education

☒ gender

☒ income

☒ resid

Equation: UNTITLED Workfile: UNTITLED::Untitled\

View Proc Object Print Name Freeze Estimate Forecast Stats Resids

Dependent Variable: INCOME

Method: Least Squares (Gauss-Newton / Marquardt steps)

Date: 10/15/21 Time: 18:18

Sample: 1 32

Included observations: 32

INCOME=C(1)+C(2)\*EDUCATION+C(3)\*GENDER

|      | Coefficient | Std. Error | t-Statistic | Prob.  |
|------|-------------|------------|-------------|--------|
| C(1) | 17.80774    | 5.509647   | 3.232102    | 0.0031 |
| C(2) | 2.585757    | 0.314338   | 8.226030    | 0.0000 |
| C(3) | -9.667076   | 2.720810   | -3.553014   | 0.0013 |

R-squared 0.743388

Adjusted R-squared 0.725690

S.E. of regression 7.622096

Sum squared resid 1684.794

Log likelihood -108.8246

F-statistic 42.00551

Prob(F-statistic) 0.000000

Mean dependent var 53.74213

S.D. dependent var 14.55305

Akaike info criterion 6.989040

Schwarz criterion 7.126452

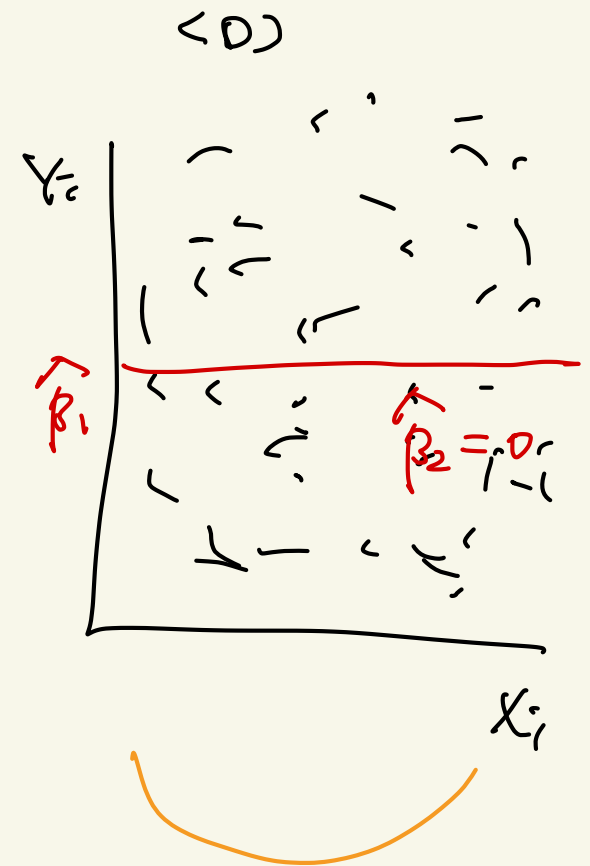
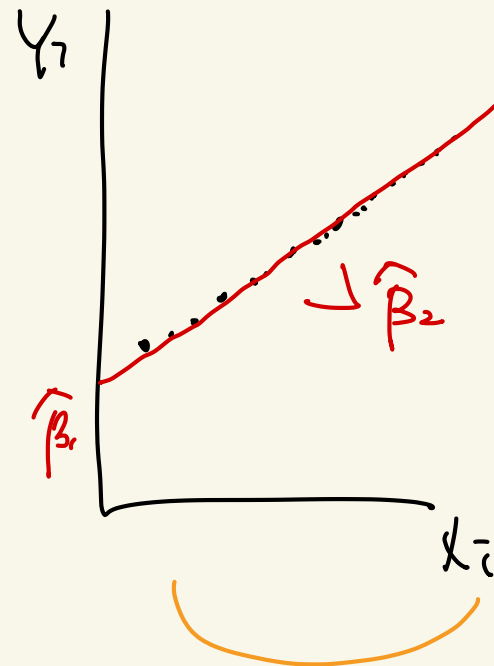
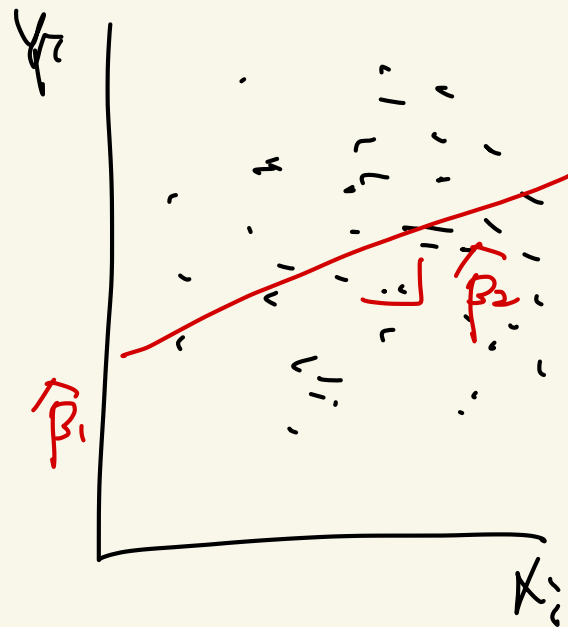
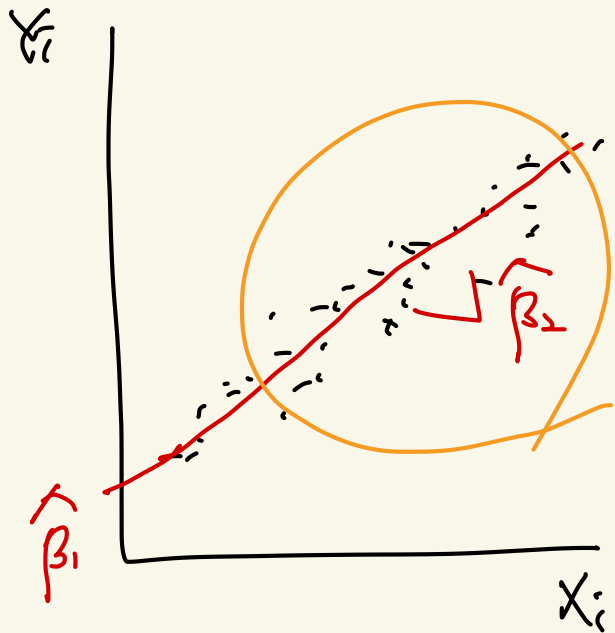
Hannan-Quinn criter 7.034588

Durbin-Watson stat 1.983339

all coef=.

$$\text{Q1)} \quad Y_i = \beta_1 + \beta_2 X_{2i} + e_i$$

<A>  $R^2$  > <B>  $R^2$  <C>



$$0 < R^2 < 1$$

$$R^2 = 1$$

$$R^2 = 0$$

$\Rightarrow \text{case C} > \text{case A} > \text{case B} > \text{case D}$

coefficient of determination, 결정계수

# 1. $R^2$ (R - square)

○  $X_1$ 의 변화가  $Y_2$ 의 변화  
 크기를 알려 주는 척도  $R^2$

- 표본회귀방정식

$$Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_{2i} + \hat{e}_i$$

이 식을 deviation from mean으로 고쳐쓰면

$$\Rightarrow (\hat{y}_i - \bar{y}) = \hat{\beta}_2 (\hat{x}_{2i} - \bar{x}) + \hat{e}_i$$

양변을 제곱하여 더하면  
 $(\hat{y}_i)^2 = (\hat{\beta}_2 \hat{x}_{2i} + \hat{e}_i)^2$

$$\sum \hat{y}_i^2 = \underbrace{\sum \hat{\beta}_2^2 \hat{x}_{2i}^2}_{\sum \hat{y}_i^2} + \sum \hat{e}_i^2 + 2 \cdot \hat{\beta}_2 \underbrace{\sum \hat{x}_{2i} \hat{e}_i}_{=0} \quad \left( \begin{array}{l} \because \text{by f.o.c} \\ \sum \hat{x}_{2i} \hat{e}_i = 0 \end{array} \right)$$

# 1. $R^2$ (R - square)

$$\sum_{i=1}^n y_i^2 = \sum_{i=1}^n \hat{y}_i^2 + \sum_{i=1}^n \hat{e}_i^2$$

양변을  $n$ 으로 나누면

$$\begin{array}{ccc} \text{T.S.S} & \text{ESS} & \text{RSS} \\ \frac{1}{n} \sum y_i^2 & = \frac{1}{n} \sum \hat{y}_i^2 & + \frac{1}{n} \sum \hat{e}_i^2 \\ \text{total sum of } y_i & \text{Explained sum of } y_i & \text{Residual sum of } y_i \end{array}$$

$\frac{\sum_{i=1}^n y_i^2}{n}$ 은  $y$ 의 분산의 추정치라고 볼 수 있다.

# 1. $R^2$ (R - square)

$$\frac{\sum_{i=1}^n y_i^2}{n} = \frac{\sum_{i=1}^n \hat{y}_i^2}{n} + \frac{\sum_{i=1}^n \hat{e}_i^2}{n}$$

Total sum of squares = explained sum of squares + residual sum of squares

$$\text{TSS} = \text{ESS} + \text{RSS}$$

"Definition"

(그래프)

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

• 만약,  $X_i$ 가  $Y$ 와  $\approx$  관계가 강해지면  $\uparrow$ ,  $\underline{\underline{R^2 \uparrow}}$

$R^2 \uparrow$  것이 항상 best? No!

## 2. adjusted $R^2$

- $R^2$ 의 한계: 설명변수를 추가할 수록 커짐

예)  $Y_i = \hat{\beta}_1^* + \hat{\beta}_2^* X_{2i} + \hat{e}_i^*$  — ①  $\Rightarrow R^2 = 0.8$

$Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 X_{3i} + \hat{e}_i$  — ②  $R^2 = 0.805$  ✓

불필요한.

$X_{3i}$ 가 불필요한 변수임에도 불구하고

$\downarrow$   
 $\rightarrow \sum \hat{y}_i^2 \uparrow$   
 $\Leftrightarrow \sum \hat{e}_i^2 \downarrow$   
 $\Rightarrow \textcircled{2} \quad R^2 \uparrow$

$$\underbrace{\sum_{i=1}^n \hat{e}_i^{*2}} \geq \underbrace{\sum_{i=1}^n \hat{e}_i^2}$$

## 2. adjusted $R^2$

penalty 부여.

- adjusted  $R^2$  : 설명변수를 많이 포함시킨 것에 대해 일정한 패널티를 부여

$$R^2 = 1 - \frac{\sum \widehat{e}_i^2}{\sum y_i^2}$$

$$\rightarrow \overline{R^2} = 1 - \frac{\sum \widehat{e}_i^2 / (n-k)}{\sum y_i^2 / (n-1)}$$

$\Rightarrow$  추가된 설명변수  
효율성  $\downarrow$   
불순도  $\uparrow$

"Principle of parsimony"

$\overline{R^2}$ 의 원리.

①  $k \uparrow$  복잡도

$\rightarrow$  ①  $\overline{R^2} \downarrow$  작아짐 ( $\because$  penalty)

$\rightarrow$  ②  $R^2 \uparrow$  (model fit 이 더

④ 증명한 변수가 추가되었을 때?

$R^2 \uparrow$  (④의 효율성  
증명되었는지 여부)

⑤ 불필요한 변수?  
( $\overline{R^2} \downarrow$ )



### 3. $R^2$ 와 F-test

- 모집단 회귀식 :  $Y_i = \beta_1 + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + e_i$  ✓
- 귀무가설 :  $H_0: \beta_2 = \beta_3 = \dots = \beta_k = 0$  ✓  
 → If  $H_0$  true, "0"이 가장다.
- ✓ Unrestricted regression :  $Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_{2i} + \dots + \hat{\beta}_k X_{ki} + \hat{e}_i$
- ✓ Restricted regression :  $Y_i = \hat{\beta}_1^* + \hat{e}_i^*$

$$R^2 = 1 - \frac{\sum \hat{e}_i^2}{\sum y_i^2}$$

$\Rightarrow R^2 \leq 0$  if  $H_0$  true. // If  $H_0$  true,  $\sum \hat{e}_i^2 \approx \sum \hat{e}_i^{*2}$   
 $R^2 \gg 0$ , we reject  $H_0$ .  
 test-stat?? No! ( $\because$  unknown probability)

### 3. $R^2$ 와 F-test

검정 통계량의 분포?  $\left\{ \begin{array}{l} 계산? \\ 분포? \end{array} \right.$

F-test

$$\frac{F - \text{test}}{F - \text{value}} \sim F(\underbrace{J, n - \text{자유도}}_{\text{모수 개수 } (k-1)})$$

- $f - \text{value} = \frac{(\sum_{i=1}^n \hat{e}_i^{*2} - \sum_{i=1}^n \hat{e}_i^2) / (k-1)}{\sum_{i=1}^n \hat{e}_i^2 / (n-k)} \sim F(k-1, n-k)$

- Unrestricted Reg.  $\boxed{Y_i = \hat{\beta}_1 + \dots + \hat{\beta}_k X_{ki} + \hat{e}_i}$   
 $\Rightarrow \sum \hat{e}_i^2$

e.g.  $Y_i = \mu + e_i$

- Restricted Reg.  $\boxed{Y_i^* = \hat{\beta}_1^* + \hat{e}_i^*}$

$\Rightarrow \sum \hat{e}_i^{*2}$

$Y_i = \bar{Y} + \hat{e}_i^*$   
 $\hat{e}_i^* = Y_i - \bar{Y} = y_i$

$$F = \frac{(\sum \hat{e}_i^{*2} - \sum \hat{e}_i^2) / k-1}{\sum \hat{e}_i^2 / n-k} \sim F(k-1, n-k)$$

$$= \frac{(\cancel{\sum y_i^2} - \sum \hat{e}_i^2) / k-1}{\frac{\sum \hat{e}_i^2}{\cancel{\sum y_i^2}} / n-k} = \frac{(1 - \frac{\sum \hat{e}_i^2}{\sum y_i^2}) / k-1}{\frac{\sum \hat{e}_i^2}{\sum y_i^2} / n-k}$$

### 3. $R^2$ 와 F-test

- $f - value = \frac{(\sum_{i=1}^n \hat{e}_i^{*2} - \sum_{i=1}^n \hat{e}_i^2)/(k-1)}{\sum_{i=1}^n \hat{e}_i^2/(n-k)} \sim F(k-1, n-k)$

$$\frac{R^2 / (k-1)}{(1-R^2) / (n-k)} \sim F(k-1, n-k)$$